

Version Control Systems

An Overview of Dropbox

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Algorithm Basics and BW

Recap: Three Approaches to Version Control

Tool	Approach	Best For
GitHub	Manual commits with messages	Code, data, structured projects
Google Docs	Automatic tracking	Collaborative documents
Dropbox	Automatic file versioning	Any file type, local files

Today: Version control that works with ANY file type!

Why Dropbox for Version Control?

- ▶ Works with ALL file types (Word , Excel , PDF , images , etc.)
- ▶ Automatic backup of your local files
- ▶ Syncs  across devices
- ▶ Version history for 30 days (or more with paid plans)
- ▶ Familiar file system interface
- ▶ No special software needed
- ▶ Works with files created in other applications

Key advantage: Version control without changing how you work

When to Use Dropbox Version Control

Ideal for:

- ▶  Excel spreadsheets with critical data
- ▶  PowerPoint presentations that evolve
- ▶  Word documents (when not collaborating in real-time)
- ▶  PDF files that get updated
- ▶  Image files that get edited
- ▶ Any file you work on locally

When to Use Dropbox Version Control

Less ideal for:

- ▶ Real-time collaboration (use  Google Docs instead)
- ▶  Code projects requiring explicit version control (use  GitHub)

How Dropbox Version History Works

Automatic process:

1.  You save a file in your  Dropbox folder (or  upload to web)
2.  Dropbox automatically creates a version
3. Every time you save changes, new version is created
4. Versions are kept for **30 days** (free account)
5. You can restore  any previous version

No action needed - just save your files normally!

Note: Similar to  Google Docs but works with files from any application

Today's Hands-On Project

Scenario: Managing a project budget spreadsheet that changes frequently

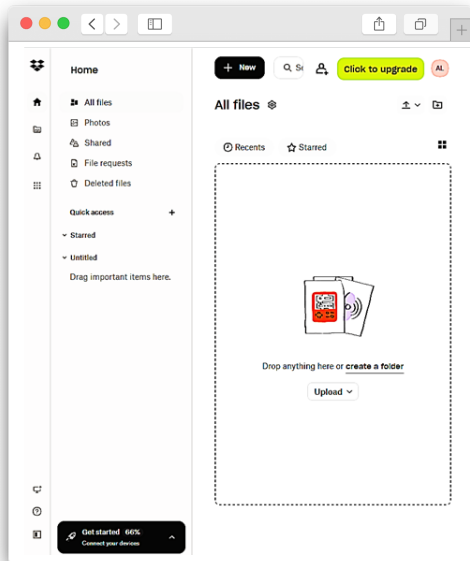
What we'll do:

1. Set up  Dropbox (if you don't have it)
2. Create an  Excel/CSV file with budget data
3. Make multiple updates simulating real project changes
4. Access version history
5. Compare different versions
6. Restore a previous version
7. Explore version control with other file types

Step 1: Set Up Dropbox

Option A: If you have Dropbox:

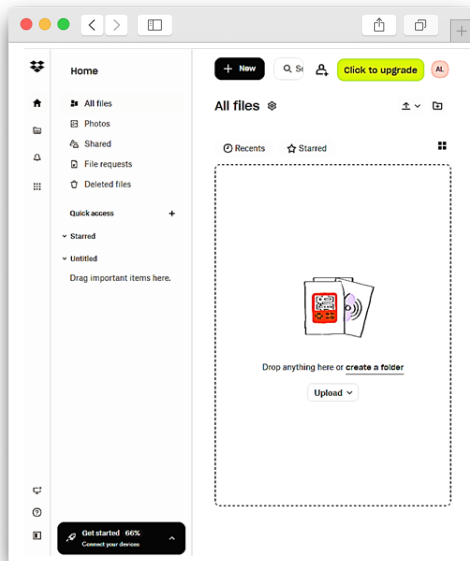
- ▶ Go to `https://www.dropbox.com`
- ▶ Sign in to your account
- ▶ Skip to Step 2



Step 1: Set Up Dropbox

Option B: Create account:

1. Go to <https://www.dropbox.com>
2. Click “Sign up”
3. Enter your name, email, password
4. Verify your email
5. Choose “Basic” (free) plan
6. Use web interface (no installation)



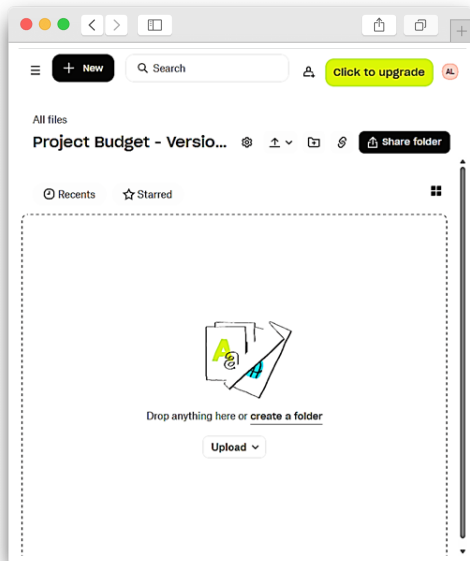
Step 2: Create a Project Folder

Organize your work:

1. In  Dropbox web interface, click “Create” → “Folder”
2. Name it:
Project Budget - Version Control Demo
3. Click “Create” and Open

Why create a folder?

- ▶ Keeps related files organized
- ▶ All files in this folder will have version control
- ▶ Mirrors real project organization



Step 3: Create Your First File

We'll create a simple CSV file (works in any spreadsheet  app):

1. Click "Create" → "Text file"
2. Name it: `budget.csv`
3. Add the following content (comma-separated, no spaces):

CSV Content

```
Category,January,February,March  
Salaries,25000,25000,25000  
Software,5000,5000,5000  
Marketing,8000,8000,8000  
Travel,3000,3000,3000  
Total,41000,41000,41000
```

4. Click "Save"

Alternative: Upload an Excel File

If you prefer working with Excel :

1. Create a simple budget spreadsheet on your computer with:
 - ▶ Columns: Category, January, February, March
 - ▶ Rows: Salaries, Software, Marketing, Travel, Total
 - ▶ Add sample numbers
2. Save  as `budget.xlsx`
3. Click “Upload” in  Dropbox
4. Drag and drop (or browse) to upload  your file

For this demo, CSV or Excel both work!

Step 4: First Update - Increase Budget

Let's simulate a budget revision:

1. Click on your budget.csv file in  Dropbox
2. Download, edit locally, and re-upload
(or edit in  Dropbox if available)
3. Change February Marketing from 8000 to 10000
4. Change February Total from 41000 to 43000
5. Save  the file

Dropbox automatically created a new version!

Note: For real work, you'd edit files locally and they sync automatically

Step 5: Second Update - Add New Category

Business needs change - add a new expense:

1. Edit your file again
2. Add a new row before "Total":

```
Equipment,2000,2000,2000
```

3. Update the Total row:

```
Total,43000,45000,43000
```

4. Save  the file

Another automatic version created!

Step 6: Third Update - Simulate Error

Let's make a mistake we'll need to fix:

1. Edit your file again
2. Delete the entire Salaries row
3. Change the Total for January to 18000 (removing salaries)
4. Save  the file

We accidentally removed a critical budget item.

This is where version control saves the day!

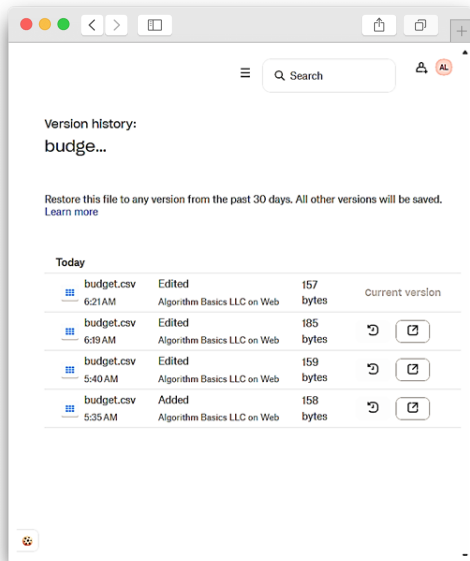
Step 7: Access Version History

Let's view all our file versions:

1. In  Dropbox, find budget.csv file
2. Hover over the file name
3. Click the three dots (...)
4. Select "Version history"
5. A panel opens showing all versions

You should see:

- ▶ Each version listed chronologically
- ▶ Timestamp for each save
- ▶ File size (may change)



Understanding the Version History Interface

What you see in version history:

- ▶ **Top version:** Current file (most recent)
- ▶ **Earlier versions:** Listed in reverse chronological order
- ▶ **Timestamps:** Exact date and time of each save
- ▶ **Actions available:**
 - ▶ Preview (preview button or the eye  icon)
 - ▶ Download  specific version
 - ▶ Restore  version

Key difference from  Google Docs: Each save creates a distinct version

Step 8: Preview Previous Versions

Examine your file history:

1. Click on the earliest version (your original file)
2. Click  "Preview" or  download to view it
3. You should see your original budget without Equipment
4. Go back to version history
5. Click on the second version
6. Notice the Marketing budget increase
7. Check the third version with Equipment added

Compare: You can see exactly how your file evolved!

Step 9: Restore a Previous Version

Let's recover the Salaries data we deleted:

1. In version history, find the version before you deleted Salaries
2. This should be the version with Equipment but before deletion
3. Click on that version
4. Click "Restore" button
5. Confirm the restoration
6. The file version history closes
7. Open your file - Salaries row is back!

Important: The restore doesn't delete the newer version - it creates a new version with the old content

What Happened Behind the Scenes

This means:

- ▶ You never truly lose data
- ▶ You can restore from a restore if needed
- ▶ Complete audit trail of all changes
- ▶ Can compare any two versions

Step 10: Try with Other File Types

Version control works with ANY file type in  Dropbox:

1. Create or upload a Word document
2. Create or upload a PowerPoint
3. Upload an image file
4. Make edits to each
5. Save/upload new versions and check version history

Try this:

- ▶ Edit a Word doc, save, edit again, save
- ▶ Check version history
- ▶ Restore to the first version

Version History Time Limits

Free Dropbox accounts:

- ▶ Keep versions for 30 days
- ▶ After 30 days, old versions are deleted
- ▶ Current version always kept

Paid Dropbox accounts:

- ▶ Extended version history (180 days to unlimited)
- ▶ Can recover deleted files for longer
- ▶ Better for business/critical data

Best practice: Don't rely solely on  Dropbox for permanent archives

Practice Exercise

Complete hands-on practice:

1. Create a new Excel or CSV file with project timeline data
2. Make at least 5 saves with different content:
 - ▶ Add new tasks
 - ▶ Change dates
 - ▶ Modify resource allocations
 - ▶ Add new columns
 - ▶ Delete and restore data
3. Explore version history
4. Download 2 different versions
5. Restore to a middle version (not latest or oldest)

Bonus: Try with a PowerPoint or Word document

Comparing All Three Tools

Feature	GitHub	Google Docs	Dropbox
File types	Text-based	Documents	Any file type
Tracking	Manual commits	Automatic	Automatic
Messages	Required	None	None
Granularity	High (you control)	Very high	Medium
Collaboration	Async reviews	Real-time	File sharing
History length	Forever	Forever	30 days (free)
Best use	Code, data projects	Collaborative writing	Local file backup

Choosing the Right Tool

Use GitHub when:

- ▶ Working with  code or structured data with explicit version labeling

Use Google Docs when:

- ▶ Writing  documents collaboratively with real-time text-heavy editing

Use Dropbox when:

- ▶ Working with diverse file types ( Excel,  PowerPoint,  images)
- ▶ Want automatic backup of local files
- ▶ Need simple file versioning without changing workflow

Real-World Business Scenarios

Scenario 1: Financial Report

- ▶ **Use:**  Dropbox + Excel
- ▶ **Why:** Version control for spreadsheet, easy to restore if formulas break

Scenario 2: Marketing Copy

- ▶ **Use:**  Google Docs
- ▶ **Why:** Multiple writers collaborating, track all edits

Real-World Business Scenarios

Scenario 3: Data Analysis Project

- ▶ **Use:**  GitHub
- ▶ **Why:** Code and data files, need explicit version control

Scenario 4: Client Presentation

- ▶ **Use:**  Dropbox + PowerPoint
- ▶ **Why:** Version history for design iterations

Questions and Discussion